

Grupo I

$$1) \alpha(0, 1, 1) \cdot \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = 1$$

✓

$$3B_{(2,3)}^T = 3$$

$$3+1=4 \neq 3$$

(F)

$$b- A = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & -1 & 1 \end{pmatrix} \xrightarrow[\substack{OL1 \\ L_3 \leftarrow L_3 - L_1}]{ } \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & -2 & 0 \end{pmatrix} \xrightarrow[\substack{OL2 \\ L_2 \leftrightarrow L_3}]{ } \begin{pmatrix} 1 & 1 & 1 \\ 0 & -2 & 0 \\ 0 & 1 & 1 \end{pmatrix}$$

$$\xrightarrow[\substack{OL3 \\ L_2 \leftarrow -\frac{1}{2}L_2}]{ } \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix} \xrightarrow[\substack{OL4 \\ L_3 \leftarrow L_3 - L_2}]{ } \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

(V)

✓

$$c- 1 + 1 - 1 + 1 = 2$$

(V)

✓

$$d- C \cdot A(2, 2) = (-1)^4 \cdot \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} = 0$$

$$\text{adj } A_{(2,2)} = 0$$

$$\frac{1}{2} \cdot 0 = 0$$

✓

(V)

$$2) a- \checkmark$$

✓

$$b- F$$

✓

$$c- F$$

✓

$$d- -x + y + z = -1$$

$$(1, 0, 1)$$

$$\Rightarrow -1 + 0 + 1 = 0 \neq -1$$

Logo n̄ há os valores pois nunca é solução da 1ª eqção

Ⓕ ✓

3) a - ✓

b - $f(1, 0, 0) = (1, 1)$

$g(0, 1, 0) = (1, 1)$

$g(0, 0, 1) = (1, 1)$

$$M_f = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \xrightarrow{\substack{L_2 \leftarrow L_2 - L_1 \\ L_3 \leftarrow L_3 - L_1}} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$\dim \text{núcleo} = n - \text{Car}(M_f) = 3 - 1 = 2$

Ⓖ

c - ✓

d - ✓

4) a - $\det(A) = 1 \times (-1) \times (-3) = 3$

Ⓕ

b - ✓

c - Se λ é autovalor de A , os de $2A + I$ são $2\lambda + 1$

Ⓖ

d - ✓

